

Variable Speed Limits (VSL): Driver Opinion Survey

Purpose: This survey aims to understand drivers' perceptions, expectations, and willingness to cooperate with dynamic speed limit systems that change based on real-time traffic conditions. All responses are anonymous and for academic use only. Participation is voluntary.

Introduction: Variable Speed Limits (VSL) — a traffic management system that changes speed limits on highways based on real-time traffic conditions (like congestion, accidents, or weather). The goal is to improve traffic flow, reduce travel time, and increase safety.

Your feedback will help us understand how drivers feel about such a system and its possible adoption in cities like Dubai.

Section 1: Background

Do you drive regularly?

- Yes
- No
- Learning / Used to

1. How often do you drive on highways (e.g. Sheikh Zayed Road)?

- Daily
- Several times a week
- Occasionally
- Rarely
- I don't drive

2. Which of the following best describes your usual driving location?

- Within a single city
- Inter-city highways
- Both equally
- Other: ____

3. Are you familiar with the concept of Variable Speed Limits (VSL)?

- Yes
- No
- Somewhat

Section 2: Opinion on VSL Implementation

Please note that extensive tests will be conducted to ensure safety, and proven to reduce travel time.

4. Would you be in favor of such a system being implemented in your city?

- Yes
- No
- Not sure

5. What benefits do you believe VSL could offer? (Select all that apply)

- Reduced congestion
- Improved safety
- Lower fuel consumption
- Better travel time estimates
- None
- Other: ____

6. What concerns do you have about VSL systems? (Select all that apply)

- Confusion due to changing limits
- Distrust in automation
- Risk of fines from sudden limit changes
- Other drivers not complying
- None
- Other: ____

7. Do you think VSL would be effective in reducing traffic jams during peak hours?

- Very effective
- Somewhat effective
- Not effective
- Not sure

Section 3: Compliance & Trust

- 8. Would you personally follow speed limits shown on an electronic VSL sign, **if legally enforced**?
- Yes
- No
- Maybe

9. Would you follow them even if **not** enforced by fines?

- Yes
- No
- Maybe

10. If field-tests demonstrate that VSL reduces accident risk and fuel use, would it change your willingness to comply?

- Yes
- No
- Maybe
- Already convinced

Section 4: General Opinions on AI

11. How familiar are you with AI-based technologies (e.g., self-driving cars, traffic prediction, automated assistants)?

- Very familiar
- Somewhat familiar
- Not familiar at all

12. What is your level of trust in AI systems used by institutions (e.g., governments, transport authorities)?

- High – I support their use to improve efficiency, automate tasks, etc.
- Moderate – I support it in low-risk areas, but not for critical decisions
- Low – I don't trust them and prefer manual human oversight

13. Experience with deployed AI systems (ChatGPT, Tesla, etc.):

- Good, I want institutions to adapt with time & tech, may have used some tools myself
- Bad, I am very skeptical, prefer humans manually do things, I refuse to give them a try

Section 5: Optional

14. Age Group:

- < 18
- 18–25
- 26–40
- 41–60
- > 60

15. What best describes your occupation sector?

- Tech/Engineering/IT
- Non-Tech (e.g., education, healthcare, business)
- Student
- Not working
- Other: _____

16. Comments or suggestions regarding VSL or traffic systems:

17. Do you consent for these choices to be stored & analysed for academic purposes only?

- Yes
- No